SYNTHETIC · SIMPLIFIED MODEL

TurbineIQ

A clear view of model versus synthetic operation.

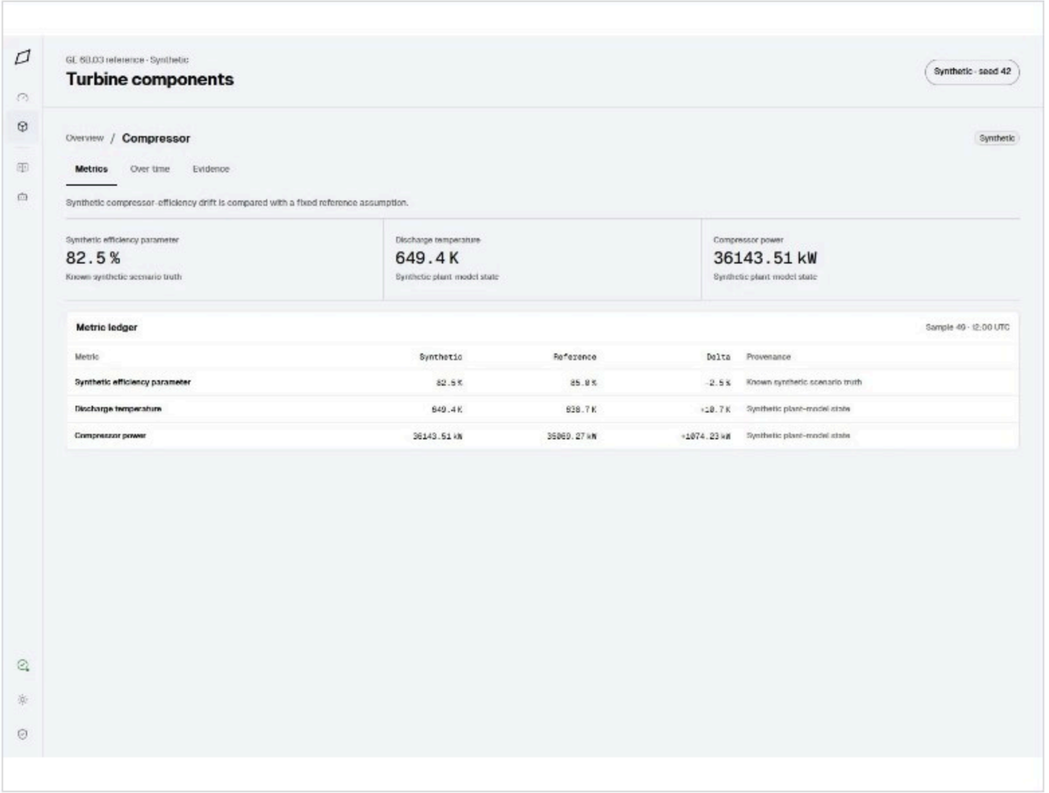
No physical connection.

A USEFUL COMPARISON

Where does synthetic data diverge from a fixed reference?

And which component helps explain the gap?

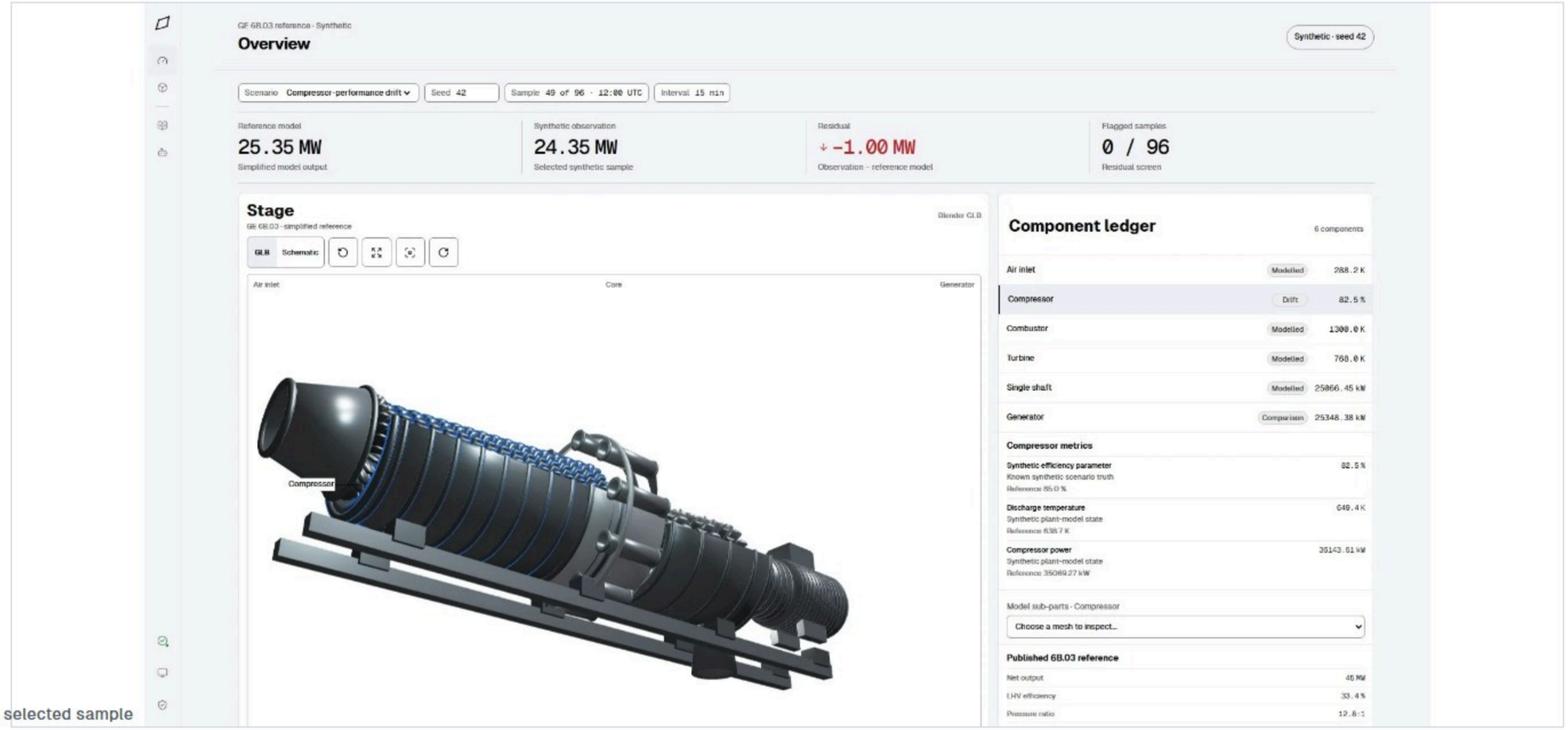
REFERENCE → OBSERVATION → COMPONENT



TWO VALUES · ONE SIGNED RESIDUAL

Reference vs observation

The model holds its assumptions fixed; the seeded synthetic sample supplies the observation.



Values remain tied to the selected sample

Seed 42

Reference model

Synthetic observation

Observation – reference

COMPRESSOR-PERFORMANCE DRIFT

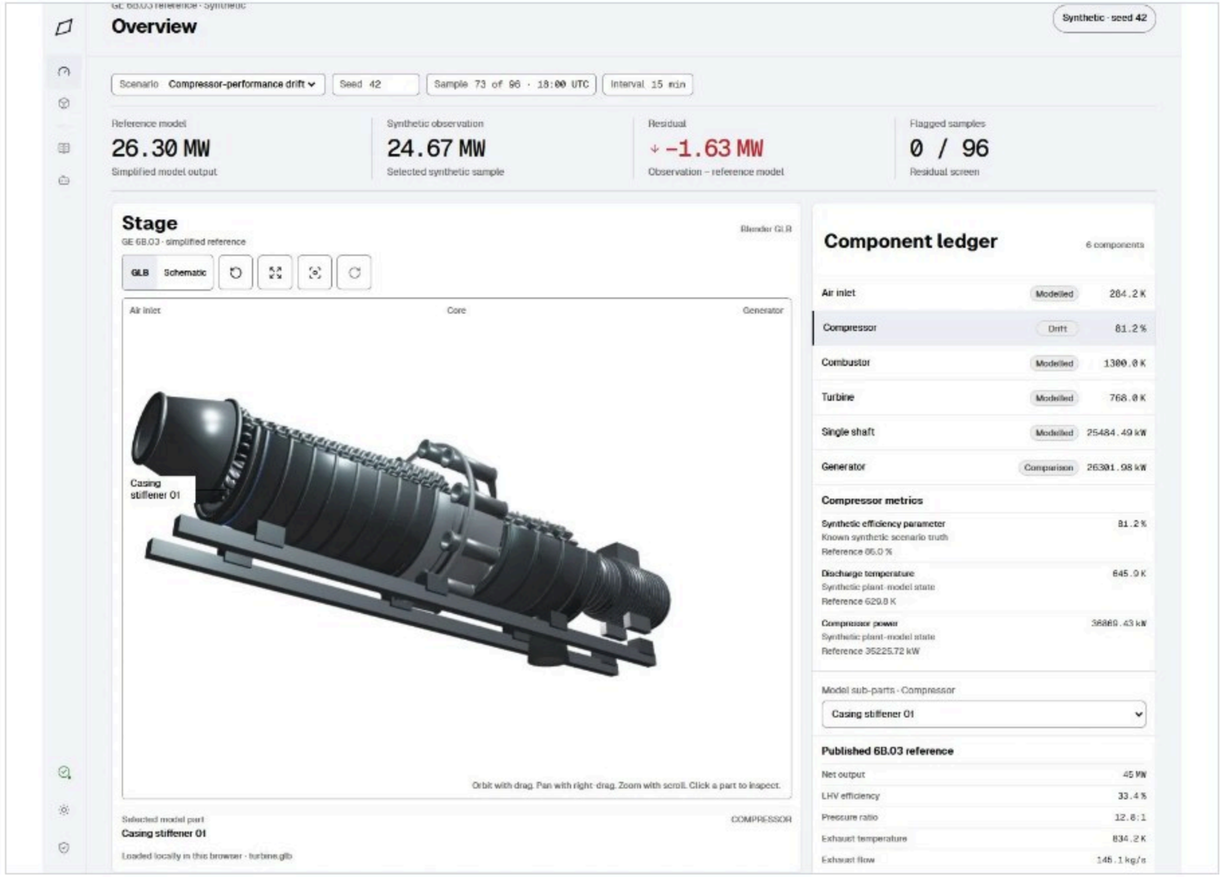
One sample, one comparison.

26.30 MW
Reference model

24.67 MW
Synthetic observation

−1.63 MW
Residual · observation minus reference

Seed 42 · sample 73 of 96 · 0 of 96 flagged



INTERACTIVE STAGE

Inspect one component in 3D.

The selected compressor mesh stays in the context of the whole model.

Selected mesh · Casing stiffener 01



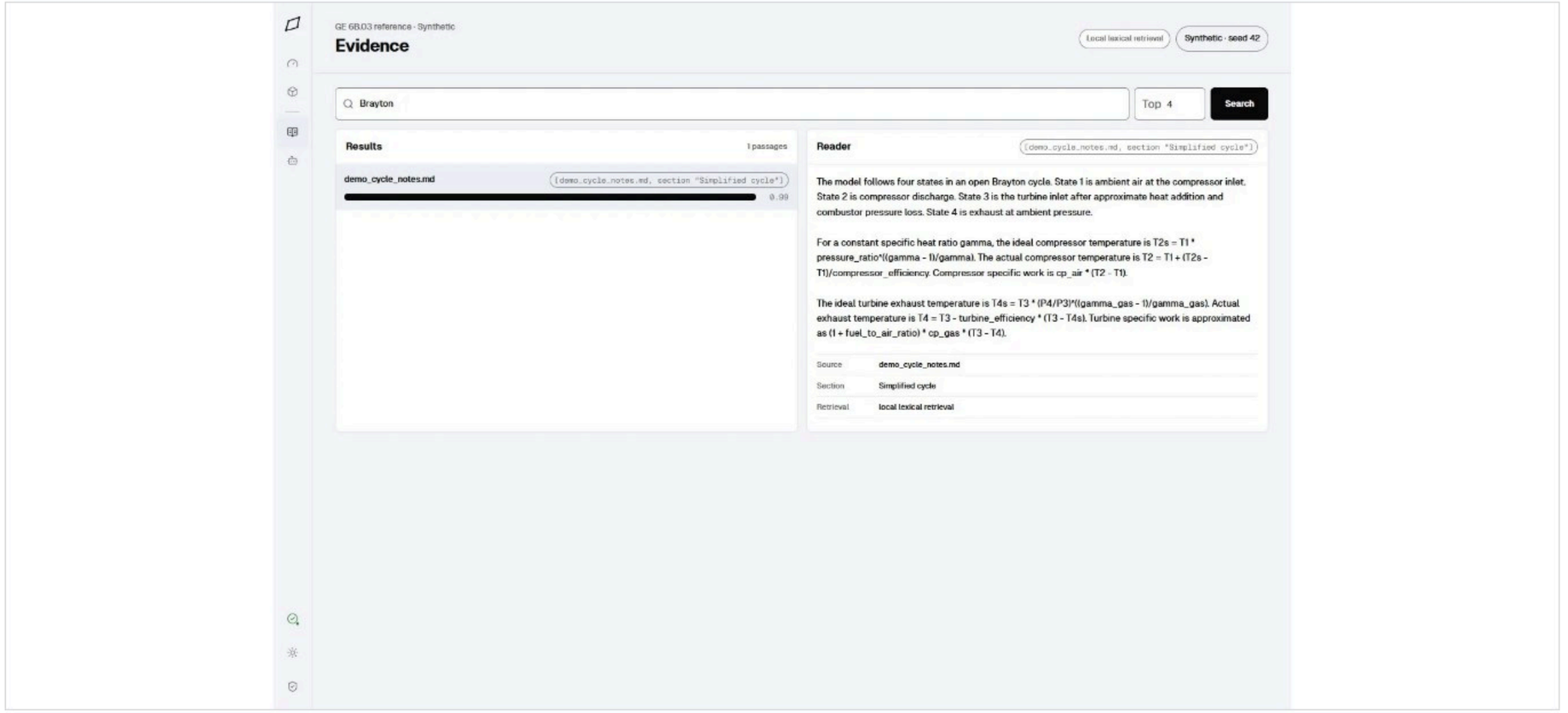
Selection · camera · isolation

Illustrative geometry

LOCAL LEXICAL RETRIEVAL

Every passage keeps its citation.

Search and inspect a source excerpt without sending it to a model.



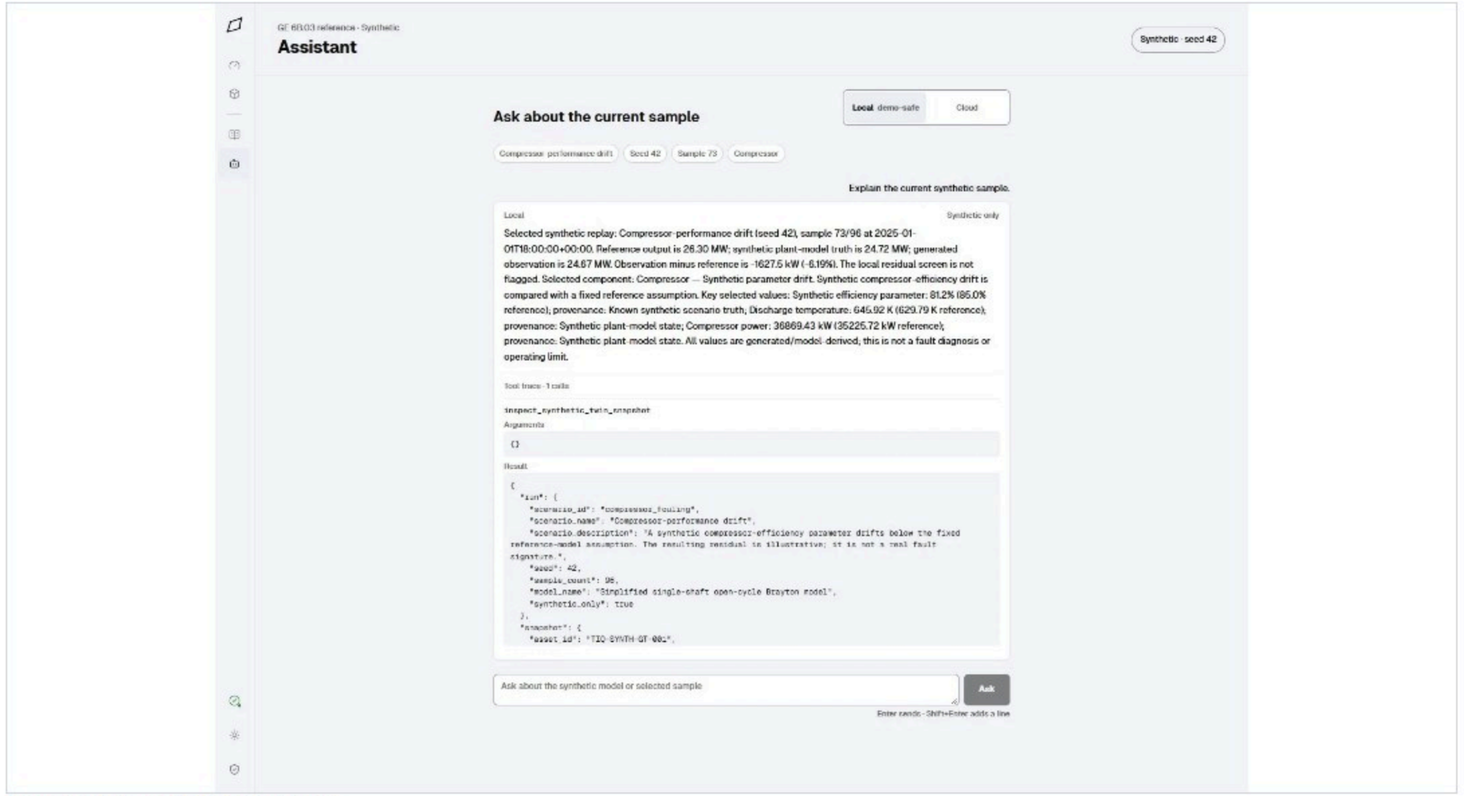
Excerpt · source · section

Plain-text rendering

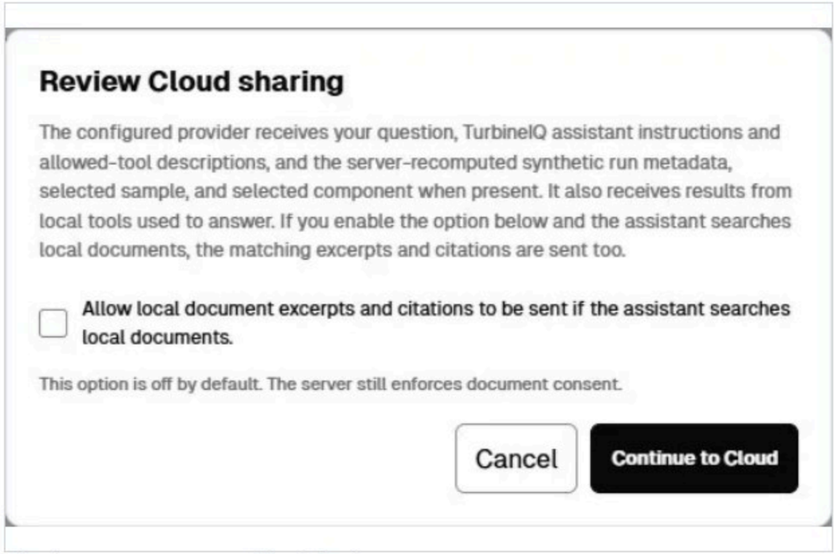
LOCAL FIRST · CLOUD BY CONSENT

Context is visible. Tool calls are inspectable.

Cloud mode opens a clear sharing sheet first; document excerpts remain a separate opt-in.



Local answer · expanded tool trace



Cloud consent · excerpts off by default

SYNTHETIC · SIMPLIFIED MODEL

Keep the boundary clear.

Synthetic data only · no real-time connection or control.

Next: review assumptions and component names with turbine engineers.